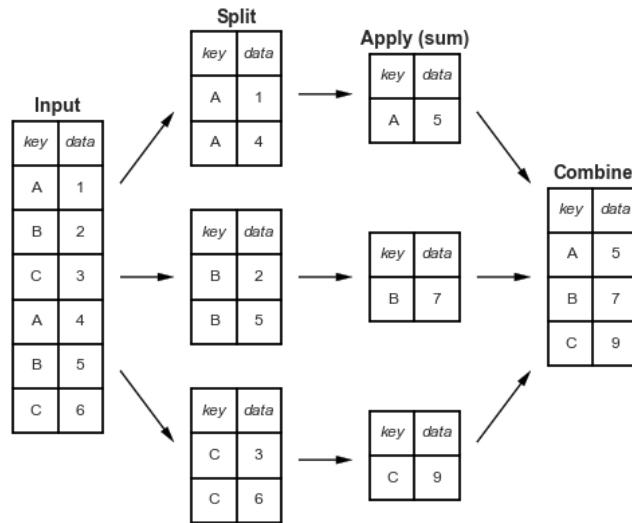


Lecture 9

Aggregating Data



Announcements

- ❖ Lab 3 today
 - Using aggregation to analyze a dataset
- ❖ Course project details coming soon
- ❖ Grades for labs will be up this week

Last Time

- ❖ Covariance & correlation coefficients

Today

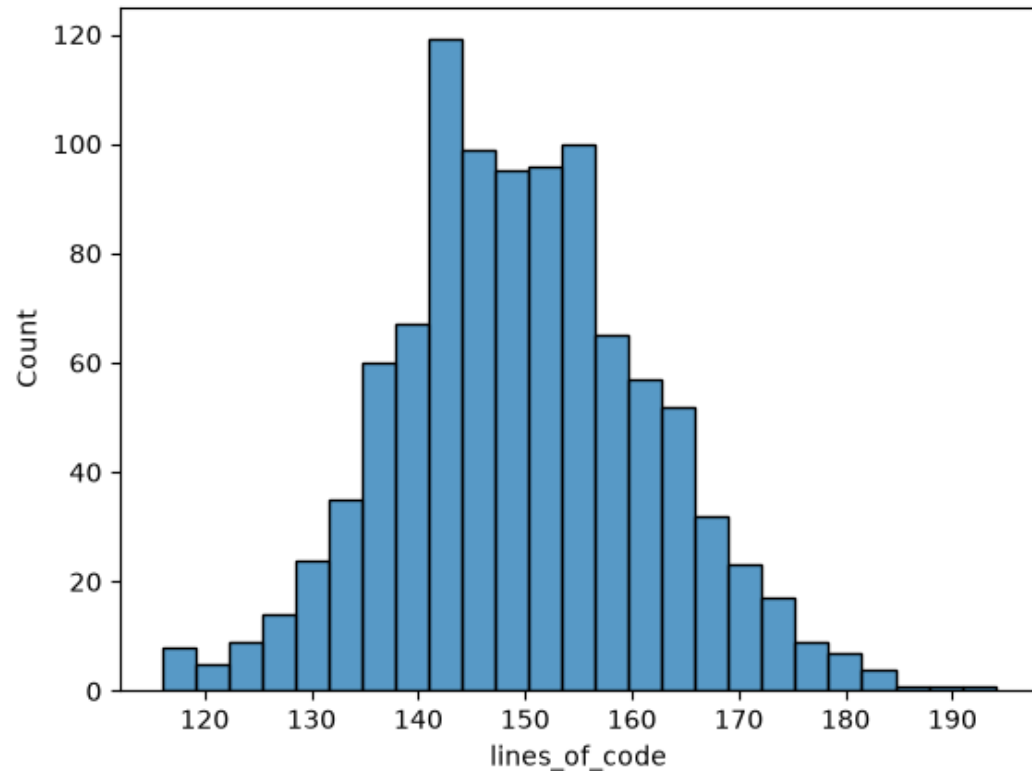
- ❖ A few more statistics
 - Mode – most common value
- ❖ Analyze debugging data

Debugging Dataset

- ❖ Describes debugging reports from students
- ❖ Let's take a look!

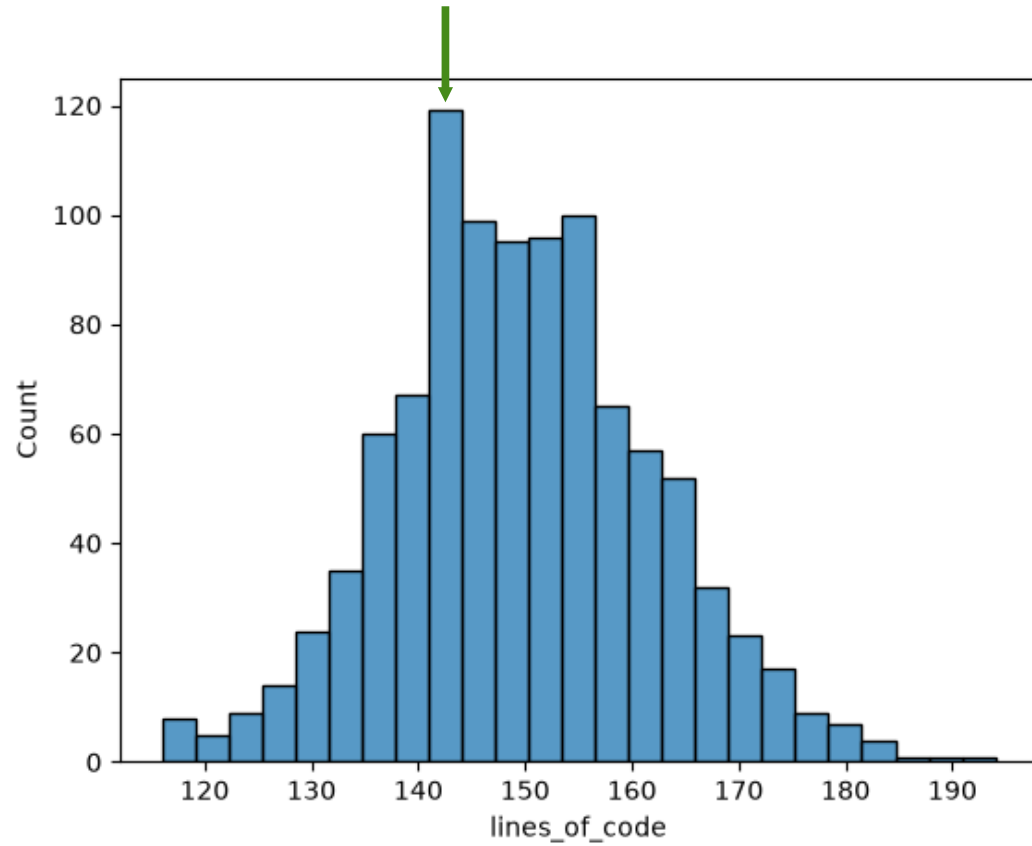
Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions



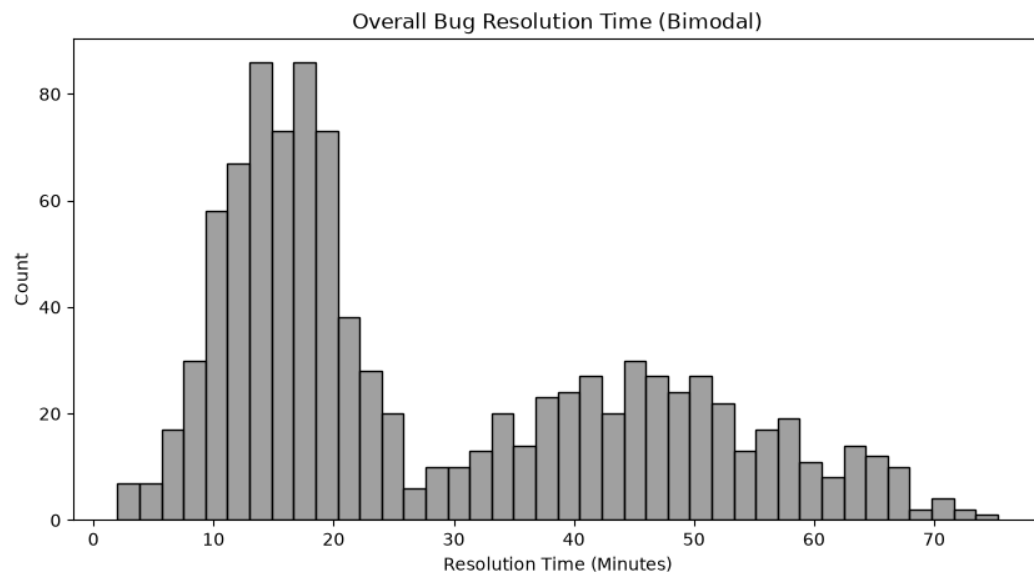
Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions



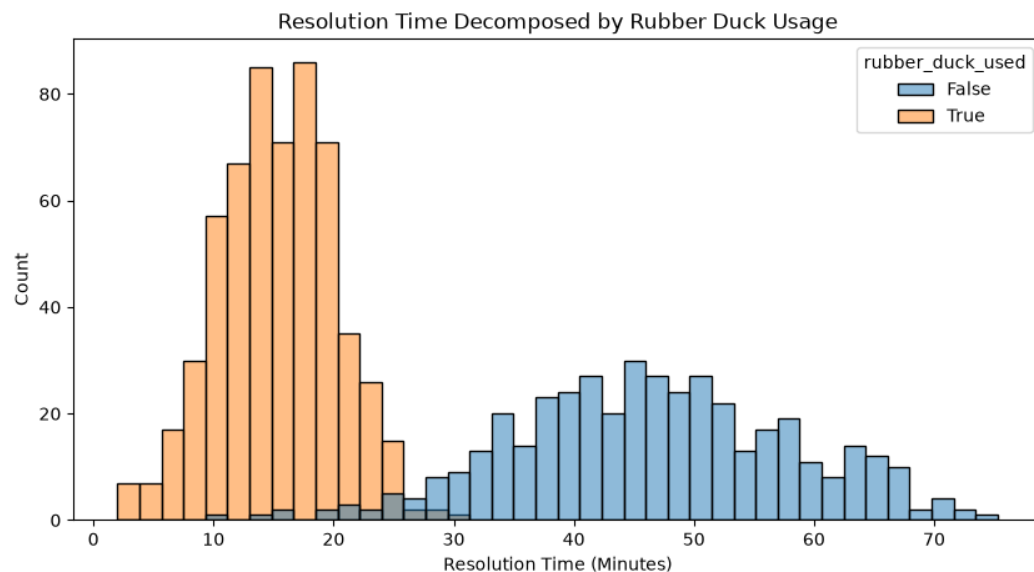
Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions
- ❖ Call this "bimodal"



Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions
- ❖ Call this "bimodal"
- ❖ Often bimodality indicates an interesting trend in the dataset



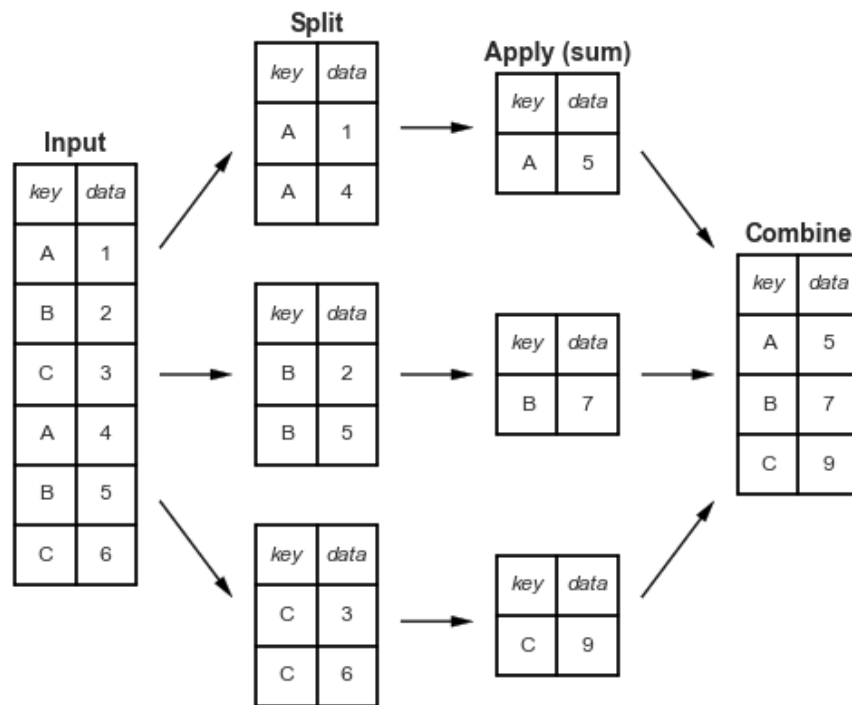
Aggregation

- ❖ Rather than computing statistics for the whole dataset, we often want to compute a statistic of each of several subgroups
- ❖ Example: average debug time by error type and debug strategy

		Rubber Duck Used	
		True	False
Syntax Error	True	4.4	32.9
	False	11.8	42.7

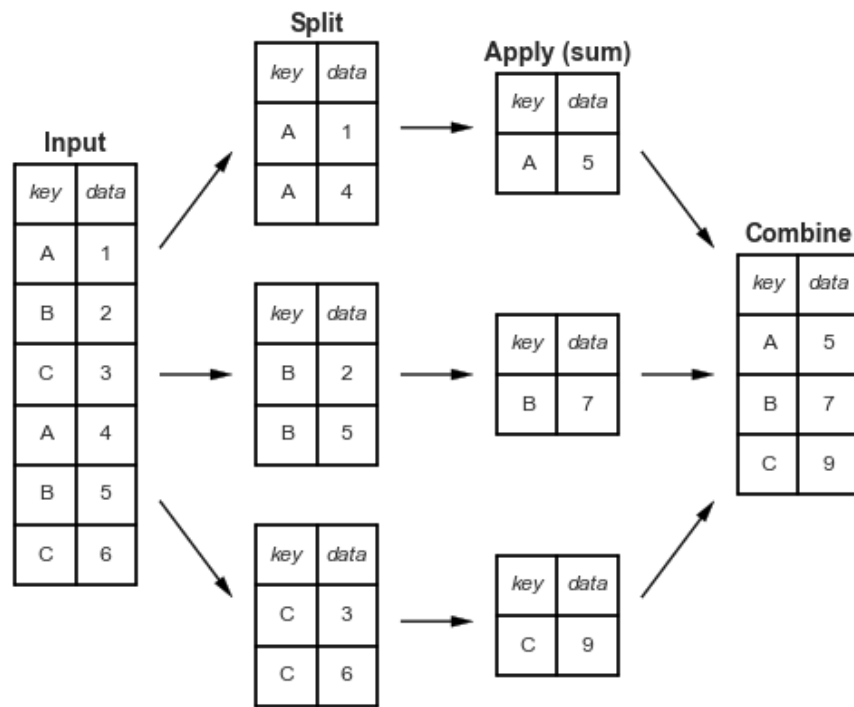
Split-Apply-Combine

- ❖ **Split:** break data into subsets based on value of *key* variables



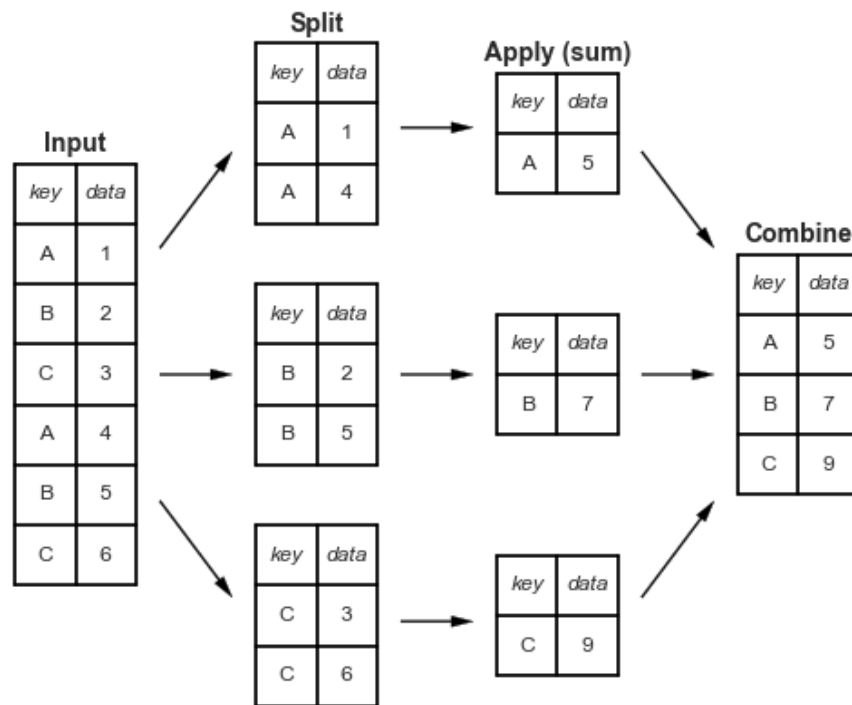
Split-Apply-Combine

- ❖ **Split:** break data into subsets based on value of *key* variables
- ❖ **Apply:** calculate statistic on each subset
 - Sum, mean, max, min, median, etc.
 - Can compute more than one statistic



Split-Apply-Combine

- ❖ **Split:** break data into subsets based on value of *key* variables
- ❖ **Apply:** calculate statistic on each subset
 - Sum, mean, max, min, median, etc.
 - Can compute more than one statistic
- ❖ **Combine:** put everything together
 - One row for each combination of key variables



Split-Apply-Combine

```
# 1. Group the data by key variables (Split)
# 2. Select the column(s) want to compute statistic on and choose the statistic (Apply/Combine)

# df.groupby(KEY_VARIABLES)['COLUMN_TO_COMPUTE_STATISTIC_ON'].AGGREGATION_FUNCTION()

grouped_mean = df.groupby(['rubber_duck_used', 'syntax_error'])['resolution_time_mins'].mean()
print(grouped_mean)
```

✓ 0.0s

rubber_duck_used	syntax_error	
False	False	42.668404
	True	32.900000
True	False	11.826897
	True	4.456944

Name: resolution_time_mins, dtype: float64

Next Time

- ❖ Read DS Ch. 3
- ❖ Aggregating Data
 - Using groupby in Python